

RFID Attendance Logger

Power and Capacity Estimation

Current draw for each operating mode and event, the reasoning behind each clock frequency, the expected number of card reads and the battery life on a 1000 mAh, 3.7 V Li-ion cell, and how the STM32L432KB's 128 kB of flash is shared between the program and the attendance log.

Hardware	Battery	Date	Status
STM32L432KB (128 kB) + ST25R3916	1000 mAh, 3.7 V nominal (3.7 Wh)	07 October 2026	Calculated, not yet measured

BATTERY LIFE

16 months

4 lectures/day, 100 students, incl. self-discharge (10 pessimistic)

IDLE SCANNING

35 μ A

Stop 2 + reader wake-up mode (57 μ A pessimistic)

PER CARD TAP

3.3 μ Ah

12.0 mC; the motor is 53 % of it

FLASH LOG

6,858 records

64 lectures of 100 students before a PC sync

1 Summary

The logger spends nearly all of its life in one of two low-power states. Switched off, it sits in Standby and draws about **2.6 μ A**. Switched on and waiting for cards, the MCU sleeps in Stop 2 while the ST25R3916 watches the antenna itself, for about **35 μ A**. The large currents (about 100 mA for the RF field and 70 mA for the vibration motor) flow only for tens of milliseconds per tap, so a whole 100-student lecture costs about **0.39 mAh** (1.45 mWh).

At four lectures a day, the device uses about **36 mAh a month**. The cell's own self-discharge (about 20 mAh a month) is a large part of the total, giving roughly **16 months per charge**, or 25 months from the device's draw alone.

The fitted MCU is the 128 kB STM32L432KB, so the program and the log share one 128 kB flash. The program takes the first 72 kB (it is 60.3 kB) and the log the last 56 kB: **6,858 records**, about 64 lectures of 100 students. The log therefore fills long before the battery empties: at four 100-student lectures a day it is full in about 16 teaching days, so the PC sync (which clears the log) sets the routine, not the charger.

Confidence. MCU figures come from the STM32L432 datasheet and are reliable. The two largest terms, the RF field current and the reader's wake-up-mode average, are estimates until the antenna is fitted and tuned. Each table gives a pessimistic column (150 mA field, 30 μ A wake-up mode, 4 false wake-ups a minute, 90 mA motor) to bound them. The flash size, the cell voltage and the program sizes were measured (section 3 and section 8).

2 Clock frequency selection

Each clock was picked for the job it does. The scanning clock matters most, because it runs on every wake-up.

Clock domain	Selection	Setting	Why
Scanning (SYSCCLK/HCLK)	MSI range 6, 4 MHz	Range 1, 0 wait states	Reset clock and the Stop 2 wake-up clock (STOPWUCK = MSI): no PLL to relock on ~1.5 wake-ups per second. The work is time-bound (5 ms field guard, 106 kbit/s frames, ADC sampling), so a faster clock only raises the current for the same window.
USB session	MSI range 9, 24 MHz	Range 1, 1 wait state	USB FS needs HCLK above 14.2 MHz (RM0394). 24 MHz is the first MSI step with margin for rendering CSV sectors on demand. Powered from VBUS, so it costs the battery nothing.
USB kernel clock	HSI48 trimmed by CRS	Locks to the host's 1 kHz SOF	Meets the USB $\pm 0.25\%$ tolerance with no crystal and no PLL. Stopped outside USB sessions.
RTC and Stop 2 timer	LSE 32.768 kHz	RTC + LPTIM1 (16-bit)	Runs through Stop 2 for well under 1 μ A. A 16-bit count at 32 768 Hz wraps every 2.0 s, so each Stop 2 sleep is capped at 1.9 s; resolution 30.5 μ s.
Reader SPI	4 MHz / 8 = 500 kHz	Mode 1, ST25R3916 max 10 MHz	Well inside the chip's limit. Register traffic during a read adds an estimated 2-3 ms of field-on time; a /2 prescaler (2 MHz) would cut it to under 1 ms.
ADC	HCLK / 4 = 1 MHz	640.5-cycle sample	0.64 ms per channel gives the 1.7 M Ω divider time to settle; the ADC is powered only for each sample.
Reader crystal	27.12 MHz	2 $\times$ 13.56 MHz carrier	Fixed by ISO 14443 (f_c = 13.56 MHz). Runs only in Ready mode or for each wake-up measurement.

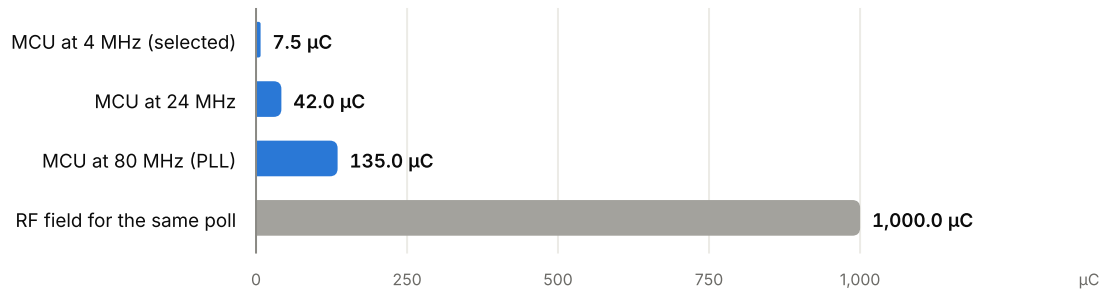
Why 4 MHz and not faster

The MCU's awake time is set by the radio and the protocol rather than by computation: a read waits out a 5 ms field guard, exchanges ISO 14443 frames at 106 kbit/s, and busy-waits on the reader's IRQ line. For a fixed window t , the charge is $Q = I_{run}(f) \times t$, so the lowest clock that meets the deadlines wins.

MCU charge during one 15 ms read window:

4 MHz	=	0.5 mA $\times$ 15 ms	=	7.5 μ C
24 MHz	=	2.8 mA $\times$ 15 ms	=	42.0 μ C
80 MHz	=	9.0 mA $\times$ 15 ms	=	135.0 μ C + PLL lock per wake-up
RF field	=	100 mA $\times$ 10 ms	=	1000 μ C

Charge for one card read: MCU clock options against the RF field



At 4 MHz the MCU is under 1 % of a read's charge. At 80 MHz it would be about 12 %, for no gain in speed.

For purely computational work (one pass of the main loop is roughly 2000 cycles), current per MHz is close to constant, so finishing faster and sleeping longer gains little: 0.5 ms at 0.5 mA (0.25 µC) at 4 MHz against 25 µs at 9 mA (0.23 µC) at 80 MHz. That small gain is lost to the PLL's start-up time and to the 4 flash wait states 80 MHz needs.

Why not slower

- **SPI with the field on.** The SPI clock is SYSCLK / 8. At 2 MHz the reader's register traffic would take twice as long while the 100 mA field is on, costing more than the MCU saves.
- **ADC sampling.** The ADC clock is HCLK / 4. A slower clock stretches the 640.5-cycle sample and keeps the ADC powered longer.
- **Low-power run (at most 2 MHz) would barely help.** The MCU is awake about 0.07 % of the time, so its run current adds only ~0.4 µA to the 35 µA idle figure.

Voltage scaling

The firmware keeps Range 1 at all clocks. Range 2 would cut run current by about 10 %, but that saves only ~0.04 µA in idle, while USB needs Range 1 anyway. Staying in Range 1 avoids switching the regulator on every USB attach. Stop 2 and Standby use the low-power regulator either way.

3 Assumptions

Battery-side currents. The 3.3 V rail comes from a linear LDO, so the battery supplies the same current the MCU, LEDs and pull-ups draw; the reader's VDD/VDD_TX and the motor sit straight on BAT+.

Quantity	Typical	Pessimistic	Source / basis
STM32L432 Stop 2 + RTC (LSE)	1.3 µA	2.0 µA	DS11451 typ., 3 V, 25 °C
STM32L432 Standby + RTC	0.5 µA	-	DS11451 typ.
STM32L432 Run, MSI 4 MHz	0.5 mA	-	DS11451 typ. (~0.4 mA at Range 2)
STM32L432 Run, 24 MHz / 80 MHz	2.8 / 9.0 mA	-	DS11451 typ., Range 1
ST25R3916 power-down	1.0 µA	-	DS12484 typ.
ST25R3916 wake-up mode, 100 ms	15 µA	30 µA	Model: ~4 µA timer + ~1.1 µC per measurement
ST25R3916 Ready (oscillator on)	4.0 mA	5.0 mA	DS12484 typ.
RF field on (from BAT+)	100 mA	150 mA	Estimate: antenna not fitted or tuned
Green / red LED (470 Ω from 3.3 V)	2.34 / 2.98 mA	3.0 / 3.9 mA	(3.3 V - Vf) / 470 Ω, Vf 2.2 / 1.9 V

Quantity	Typical	Pessimistic	Source / basis
Vibration motor (coin ERM on BAT+)	70 mA	90 mA	Assumed; measure the fitted motor
Button pull-ups (R8 10k + internal)	0.41 mA	-	Only while pressed
Battery divider R7 + R8 (7.4 M)	0.56 μ A	-	4.15 V / 7.4 M Ω
TPS7A02 LDO quiescent	25 nA	-	TPS7A02 datasheet typ.
MCP73833 leakage, no input	0.5 μ A	1.0 μ A	Assumed upper bound
False wake-ups while scanning	1 / min	4 / min	After offset learning; check dbg_nfc_false_wakes
Cell self-discharge	2 %/month	3 %/month	Typical Li-ion at room temperature
Usable capacity of 1000 mAh	900 mAh	850 mAh	Cut-off at 3.3 V, ageing margin

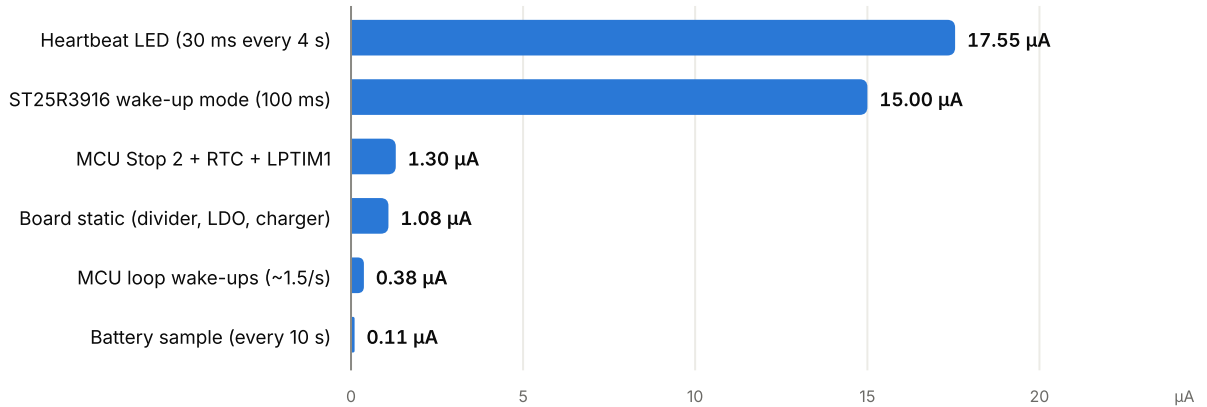
Measured on the board MEASURED 07 OCTOBER 2026

Quantity	Value	Note
Flash size (0x1FFF75E0)	128 kB	Fitted part is the STM32L432KB
Cell voltage (dbg_battery_mv)	4,152 mV	Three samples 10 s apart: 4152, 4149, 4154 mV
State of charge (dbg_battery_percent)	96 %	From the firmware's Li-ion voltage curve
ADC counts at PA7	1,876	PA7 = 3308 mV $\times$ 1876 / 4095 = 1515 mV
VDDA from VREFINT	3,308 mV	VREFINT_CAL 1652 / 1497 counts $\times$ 3000 mV
Divider check	4,153 mV	1515 mV $\times$ (4.7 M + 2.7 M) / 2.7 M

4 Operating modes

Mode	Typical	Pessimistic	What is running
Off (Standby)	2.6 μ A	3.1 μ A	MCU Standby + RTC, reader power-down, divider
Idle, scanning	35.4 μ A	56.6 μ A	Stop 2, reader wake-up mode, heartbeat
Polling mode (wake-up off)	10.1 mA	-	Reader Ready + field every 100 ms; for comparison
Card read (RF field on)	100 mA	150 mA	For ~10 ms per read; antenna untuned
Vibration motor on	70 mA	90 mA	60-250 ms per pattern
USB session	4-6 mA	-	From VBUS, not the battery (plus charge current)

Idle scanning current, 35.4 µA in total



The heartbeat LED and the reader's own antenna measurements are about 90 % of idle current. The MCU's share is about 5 %.

Heartbeat	=	$2.34 \text{ mA} \times 30 \text{ ms} / 4000 \text{ ms}$	=	$17.6 \text{ } \mu\text{A}$
Wake-ups	=	$1.5 \text{ /s} \times 0.5 \text{ ms} \times 0.5 \text{ mA}$	=	$0.38 \text{ } \mu\text{A}$
Sampling	=	$1.5 \text{ ms} \times (0.5 + 0.2) \text{ mA} / 10 \text{ s}$	=	$0.11 \text{ } \mu\text{A}$
Off	=	$0.5 + 1.0 + 0.56 + 0.025 + 0.5 \text{ } \mu\text{A}$	=	$2.58 \text{ } \mu\text{A}$
Polling mode	=	$4.0 \text{ mA Ready} + 100 \text{ mA} \times 6 \text{ ms} / 100 \text{ ms} + \text{MCU}$	=	10.1 mA

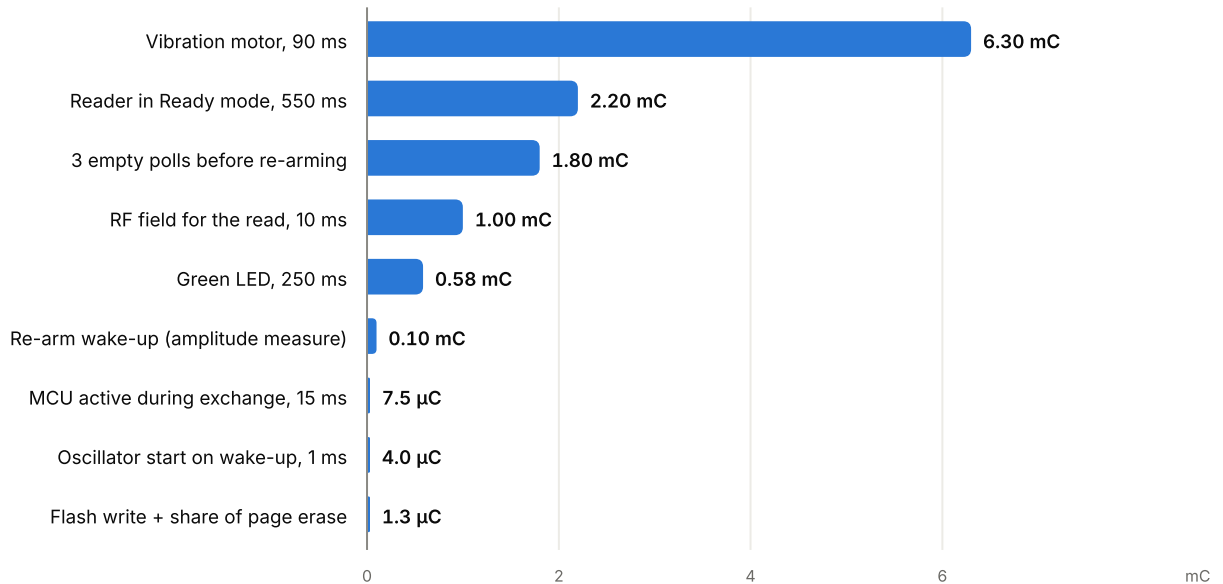
Switching the reader's wake-up mode off would take idle current from 35 µA to about 10 mA: battery life left switched on would drop from 2.9 years to 3.7 days. The wake-up mode is the most important saving in the design.

5 Charge per event

Charge = current × time for each load, summed over the event. 1 μAh = 3.6 mC.

Event	What draws current	Charge (typ.)	μAh (typ.)	μAh (pess.)
Accepted card tap	Field 10 ms, motor 90 ms, LED 250 ms, Ready 550 ms, 3 empty polls	12.00 mC	3.33	4.59
Duplicate tap	Same read, 2 × 60 ms motor + LED	13.63 mC	3.79	5.17
False wake-up	Ready ~0.3 s + 3 empty polls, no feedback	3.10 mC	0.86	1.20
Power-on (button)	Boot (20 ms at 80 MHz), reader start, 120 ms motor, 300 ms LED	9.44 mC	2.62	3.35
Lecture start (2-5 s hold)	3 s hold, 60 ms + 3 × 80 ms motor, 880 ms LED, marker write	24.30 mC	6.75	8.58
Auto power-off (3 min idle)	250 ms motor, 700 ms red LED, flush, reader power-down	19.59 mC	5.44	7.01
Power-off (5 s hold)	As above + 5 s hold + 60 ms hold buzz	25.84 mC	7.18	9.08
Short press (battery check)	0.2 s press, two 150 ms green blinks	0.78 mC	0.22	0.27
Flash page erase	22 ms at ~7 mA, once per 254 records	0.15 mC	0.04	0.04
Battery sample	1.5 ms MCU + ADC, every 10 s	1.0 μC	0.0003	0.0003

One accepted card tap, 12.0 mC (3.33 μAh)



Feedback (motor and LED) and keeping the reader awake after the read cost more than the read itself.

Motor	= 70 mA × 90 ms	= 6.30 mC
Ready mode	= 4.0 mA × (250 ms pattern + 3 × 100 ms)	= 2.20 mC
Empty polls	= 3 × 100 mA × 6 ms	= 1.80 mC
Read	= 100 mA × 10 ms (guard, REQA, anticollision, SELECT)	= 1.00 mC
LED	= 2.34 mA × 250 ms	= 0.58 mC
Total	= incl. re-arm, MCU, oscillator, flash	= 12.00 mC = 3.33 μAh

6 A lecture, and battery life

The model lecture: the lecturer powers the unit on and starts a lecture with a button hold; 100 students tap over 15 minutes, 5 of them twice; 3 minutes after the last tap the unit switches itself off. It is on for 18 minutes.

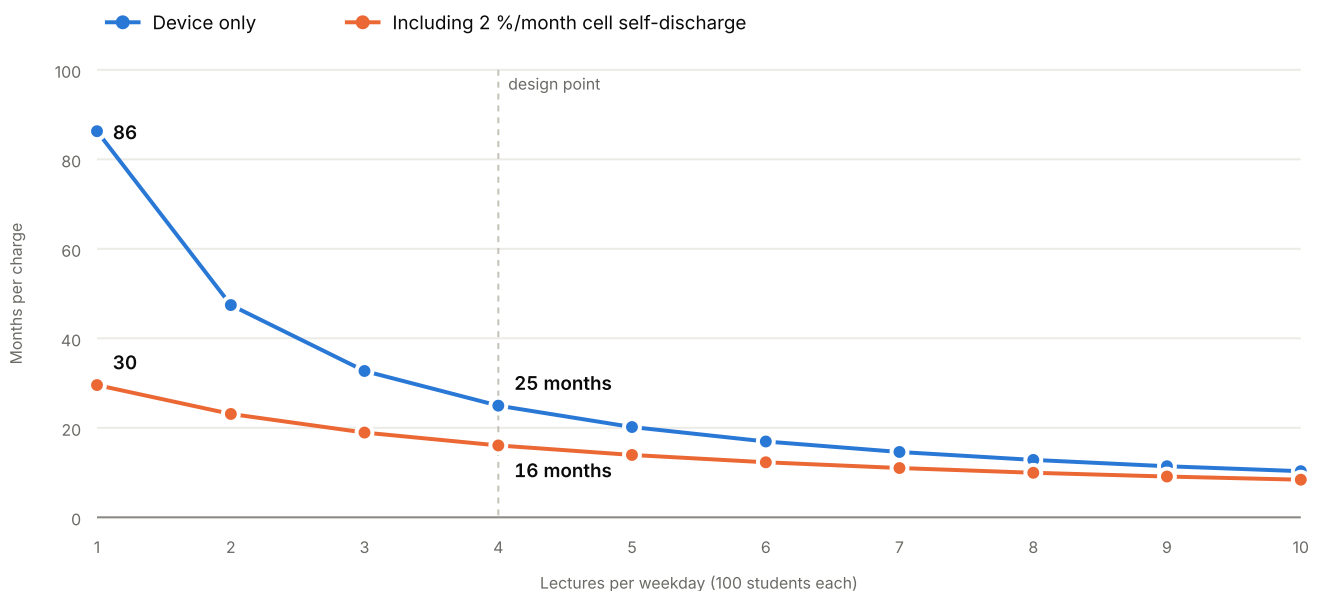
Part	μAh	Share
100 accepted taps	333.3	85 %
5 duplicate taps	18.9	5 %
False wake-ups (1/min)	15.5	4 %
Idle scanning, 18 min	10.6	3 %
Lecture start (button hold)	6.7	2 %
Auto power-off pattern	5.4	1 %
Power-on	2.6	1 %
Total per lecture	393	100 %

Per month at 4 lectures/weekday:

lectures	= 4 × 21.7	= 86.9
device	= 86.9 × 0.393 mAh	
	+ 2.58 μA × 730 h	= 36.1 mAh
self-dis.	= 2 % × 1000 mAh	= 20.0 mAh
total		= 56.1 mAh

Life = 900 mAh / 56.1 mAh = 16.1 months
(device only: 25.0 months)

Months per charge against lectures per day



Light use is limited by the cell's self-discharge, not the device. A full charge lasts over a year at up to about 4 lectures a day.

Scenario (weekdays)	µAh per lecture	Months, device only	Months, with self-discharge	Months, pessimistic
Light: 1/day × 50 students	215	137	34	21
Typical: 4/day × 100 students	393	25	16	10
Busy: 6/day × 150 students	571	12	9	6
Heavy: 8/day × 200 students	745	7	6	4

Shelf time when never switched on: 900 mAh / 2.6 µA = 40 years from the device's draw alone, but self-discharge empties the cell in roughly 3.8 years. Store the device charged and recharge it each term.

7 Card reads per charge

A tap alone costs 3.33 µAh, so a charge could fund **270,049** taps if nothing else drew current. With each lecture's idle time, power-on and power-off, duplicates and false wake-ups shared out among its students, the effective cost per student is higher, especially in small classes:

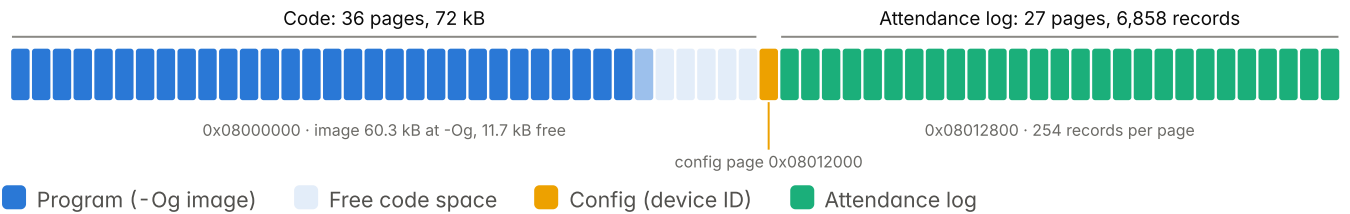
Usage pattern	µAh per student (all overheads)	Taps per charge (900 mAh)	Flash fills per charge
30 students per lecture	4.95	181,798	27x
60 students per lecture	4.20	214,044	31x
100 students per lecture	3.93	228,910	33x
200 students per lecture	3.73	241,490	35x

Even small classes get well over 150 000 reads per charge (self-discharge aside), which is 27 to 35 times the 6,858 records the flash can hold. Before the battery is a limit, the log has to be read and cleared by the PC app many times over (by #CLEARLOG when it starts a lecture).

8 Flash: program and attendance log

The fitted MCU is the **STM32L432KB**: its flash - size word at 0x1FFF75E0 reads 128 (measured on the board). Its 128 kB is 64 pages of 2 kB, erased a page at a time, and the program and the log must share them. Earlier firmware assumed the 256 kB STM32L432KC and kept its data at 0x08020000, which the KB does not specify; the layout below fits the KB.

Flash map, one block per 2 kB page

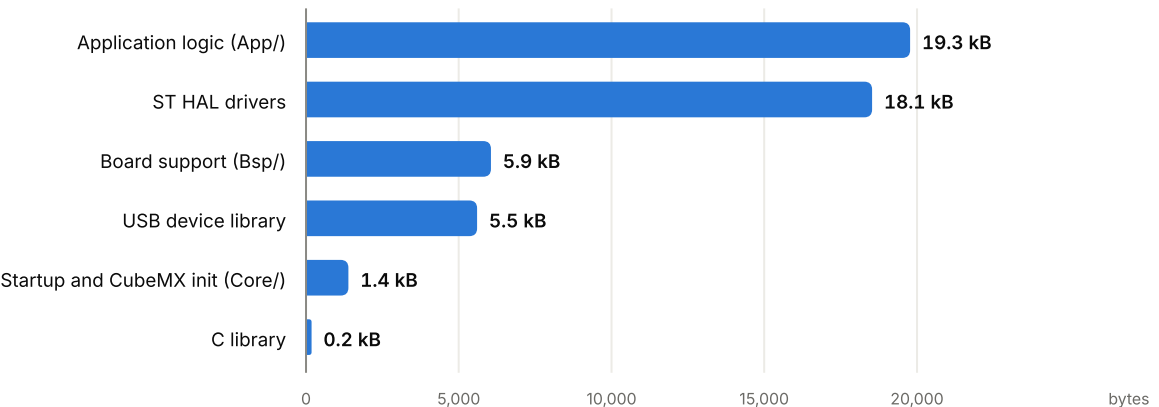


Program size

Measured with the project's own toolchain (arm-none-eabi-gcc 13.3, nano libc, unused sections removed):

Build	Image (bytes)	In 72 kB	Fits	Note
CubeIDE Debug before this change (-O0)	99,244	96.9 kB	No	Larger than any split that leaves a useful log
make and CubeIDE Debug (-Og)	61,772	11.7 kB free	Yes	84 % of the code region; full debugging
CubeIDE Release (-Os)	51,256	21.9 kB free	Yes	Shipping build
-Os with link-time optimisation	46,220	26.9 kB free	Yes	Not used: hardest to debug

Where the 50.1 kB -Os image goes



Application code and ST's HAL are about three quarters of the image. The HAL is used through its generic drivers; replacing it with direct register code could save several kB more, at the cost of portability.

Choosing the split

Every page given to code is 254 records fewer in the log. The CubelDE Debug build used to compile at -O0 (97 kB), which fits no useful split, so it now builds at -Og like the Makefile. With -Og as the largest build that must fit:

Code / data	Log pages	Records	Lectures of 100	Headroom at -Og	Headroom at -Os
64 kB / 64 kB	31	7,874	74	3.7 kB	13.9 kB
72 kB / 56 kB (chosen)	27	6,858	64	11.7 kB	21.9 kB
80 kB / 48 kB	23	5,842	55	19.7 kB	29.9 kB

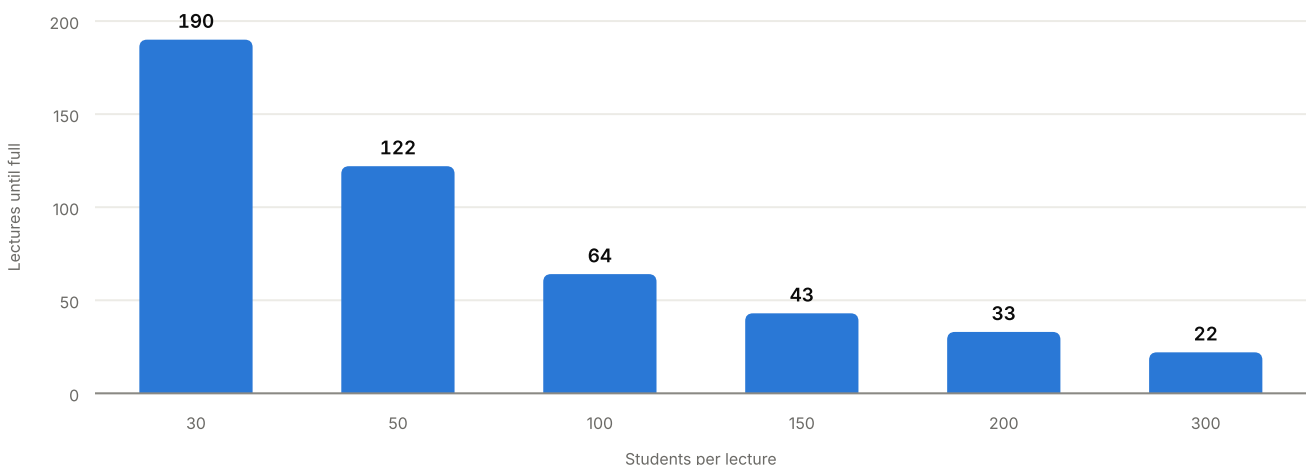
72 kB of code leaves 11.7 kB (16 %) for future features at -Og and still gives the log 6,858 records. A 64 kB code region would add 1,016 records (+15 %) but leave only 3.7 kB at -Og, forcing -Os for debugging. If records ever matter more than debugging, -Os with link-time optimisation (45.1 kB) would make the 64 kB split comfortable.

The linker script (STM32L432KBUX_FLASH.ld) gives the program 72 kB, so an image that outgrows it fails to link instead of overwriting the log. Link-time and compile-time checks keep the linker script, the board header and the log layout in step.

Log capacity

```
Page      = 2048 bytes = 256 double-words
Records   = 256 - 1 header - 1 footer           = 254 per page
Log        = 27 pages × 254                     = 6,858 records (8 bytes each: card ID + time)
Lecture    = 1 header + ceil((module + lecture + 2 NULs) / 4) text records
            = 1 + ceil((6 + 10 + 2) / 4) = 6 for "EE4020" / "Lecture 12" (range 4-16)
Lectures   = floor(6,858 / (N students + 6))
```

Lectures the log can hold before a PC sync



Duplicates are not stored, so N counts distinct students in each lecture.

Students / lecture	Records / lecture	Lectures until full	Student taps stored	At 4 lectures per weekday
30	36	190	5,700	48 days (9.5 weeks)
50	56	122	6,100	30 days (6.1 weeks)
100	106	64	6,400	16 days (3.2 weeks)
150	156	43	6,450	11 days (2.1 weeks)
200	206	33	6,600	8 days (1.6 weeks)
300	306	22	6,600	6 days (1.1 weeks)

- **Students per lecture:** up to **6,852** in a single lecture after a clear (log size minus one marker).
- **Registered students:** not limited by the device. It keeps no card list; any 32-bit card ID below 0xFFFFFFFF00 is stored, and the PC app decides who each card belongs to.
- **When the log is full** new taps are refused with the red error pattern, because APP_LOG_WRAP_WHEN_FULL is 0. Up to 128 records wait in RAM (flushed at 80 % or 5 s after the last tap) and are written before power-off.
- **USB view:** a full log is 6,858 rows × 32 bytes = 219 kB of ATTEND.CSV (429 clusters), well within FAT12's 4084-cluster limit.
- **Flash wear:** each log page is erased once per fill-and-clear cycle. At the STM32L4's 10,000-cycle endurance, that is 10,000 full logs (about 69 million records): not a limit in practice.
- **Moving a unit to this layout:** the new firmware does not read the old data at 0x08020000. Import the unit's log with the PC app before reflashing, and set its device ID again afterwards.

9 Where to save more power

Ranked by effect on the typical lecture. None of these is needed to reach a year per charge.

Change	Saving	Effect
Motor pulse 90 ms to 50 ms	-2.8 mC per tap	23 % of each tap
Re-arm wake-up right after a read (skip 3 empty polls)	-3.0 mC per tap	25 % of each tap
Heartbeat 30 ms to 10 ms	-11.7 µA idle	33 % of idle
Wake-up period 100 ms to 200 ms	-5.5 µA idle	16 % of idle; ~100 ms slower response
SPI prescaler /8 to /2	-0.14 mC per read	1 % of each tap
Voltage scaling Range 2 while scanning	-0.04 µA idle	<1 % (the MCU is awake 0.08 % of the time)

10 Verification plan

- Fit and tune the antenna, then measure the RF field current at BAT+ during a read (largest single uncertainty).
- Measure average idle current with a current-integrating meter (Nordic PPK2 or similar) over 60 s with no card present, with no debugger attached, since an attached debugger keeps the MCU out of Stop 2. Expect about 35 µA.

- Integrate one tap and compare it with 12 mC; read `dbg_nfc_false_wakes` after an hour to confirm the false wake-up rate.
- Measure Standby current with the unit switched off (expect about 2.6 μA), and the fitted motor's running current.
- Fill the log on the bench (a test that taps 6,858 IDs) and confirm the red error pattern on the next tap.
- Feed the measured values back into this model (`Firmware/docs/power_report.py`); the calculations scale linearly with each term.

All values are at room temperature. Li-ion capacity falls 10-20 % near 0 °C and with ageing, already partly covered by the 90 % usable-capacity factor.